Sea Ice Classification via Deep Neural Network Semantic Segmentation

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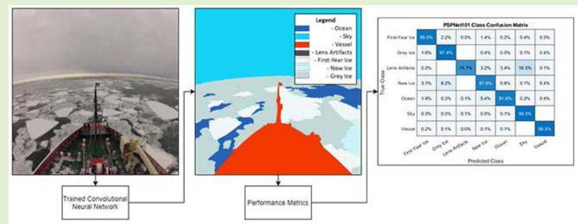
***Abstract*—Sea ice monitoring plays a critical role in any ice-breaker's journey, where standard procedures are in place to document and report sea ice types and concentration. In this paper, we propose semantic segmentation for automated detection and classification of sea ice types using camera feeds onboard an ice breaker. For this purpose, we evaluate the SegNet and PSPNet101 neural network architectures, which have proven success in navigation and mapping applications such as self-driving cars, remote sensing, and medical imagery. The networks are used to segment images based on two custom datasets, one with four classes: ice, ocean, vessel, and sky, i.e., sea ice detection dataset, and the second with eight classes: ocean, vessel, sky, lens artifacts, first-year ice, new ice, grey ice, and multiyear ice, i.e., sea ice classification dataset. The Nathaniel B. Palmer imagery, which captured 2-month footage of the icebreaker completing an Antarctic expedition was used in the creation of both datasets. A subset of the dataset was labeled to generate a 240-image training set for sea ice detection achieving an accuracy of 98% classification for the 26-image test set. The sea ice classification dataset consists of 1,090 labeled images achieving accuracies of 98.3% or greater for all ice types for the 104-image test set. These results validate the applicability of deep learning methods for sea ice detection and classification using images captured onboard an ice breaker, which can be further enhanced by incorporating additional ice types and operational data to support marine navigation and mapping applications.**

***Index Terms*—Semantic segmentation, deep neural networks, intelligent systems, machine learning.**



I. Introduction

ICE detection and classification onboard icebreakers play a critical role in any polar voyage. While this process typically relies on dedicated ice specialists to identify and report sea ice conditions, recent advances in machine vision object classification methods can offer enhanced automated workflows for sea ice monitoring activities. Object classification in machine vision systems can be broadly defined as the process of identifying types of objects seen in images. Semantic segmentation exists as a further extension of object classification. While object classification focuses on recognizing an object's existence in the frame, semantic segmentation goes a step further to label every pixel in the image with a corresponding object identifier, i.e., a class label. The class labels are selected to suit the application; for example, urban scene recognition makes use of people, sky, ocean, and cars as the class labels. Semantic segmentation is a key initial step in several state of the art intelligent system designs such as remote sensing, medical image analysis, and autonomous driving algorithms.

In this paper, we investigate semantic segmentation in the marine environment, specifically for sea ice detection and classification. Classical image processing methods have been shown to be effective under certain conditions for semantic segmentation. Adaptive thresholding, as seen in [1] presents a method for color images to be segmented on an adaptively selected threshold. Jumb *et al.* [2] propose segmentation via salient region detection. In this approach, salient regions are detected via low-level features of color and luminance and segmented accordingly. K-means clustering and watershed hybrids have also been presented for segmentation. Work in [3] demonstrates a highly accurate medical image segmentation using K-means to create initial partitions and watershed to further define and combine the segmentation. Another common approach for object segmentation is the graph cuts method [4]. This is an interactive segmentation method that takes user-defined markers to form boundaries around objects

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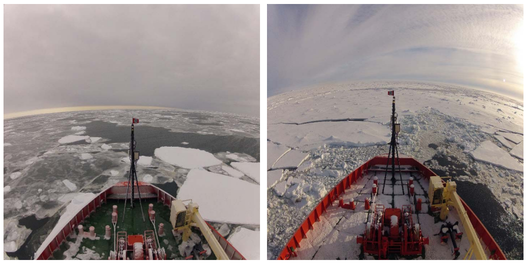

(a) (b)

Fig. 1. Examples of polar environments encountered aboard an icebreaker. (a) A grey day, with a mixed concentration of ice, taken from the Nathaniel B. Palmer dataset [31]. (b) A typical day in a first-year ice field from the same dataset.

in the image. Texture-based segmentation is another popular approach used for semantic segmentation, e.g. work in [5] presents an unsupervised color-texture region segmentation method.

While all the aforementioned image labeling techniques present a high degree of success for a particular idealized set of operating conditions, purely visual means of labeling is a difficult problem to approach. In any environment that an image recognition method is developed for, small changes in the environment can lead to degraded performance or complete failure of the system. This is the case when it comes to the harsh environment of the Arctic sea. Due to the ever-changing lighting of a typical day and glare produced off the ice and ocean combined, it is challenging to develop a visual approach to properly segment the image [6]. The variance in ice type, size, color, texture, and shape presents an additional hurdle [7], which makes a classical image processing based approach susceptible to incorrect segmentation. Furthermore, this problem is challenging simply because the Arctic has varying environmental conditions, i.e., fog, rain, low light, snow. For example both images shown in Fig. 1 feature first-year ice. However, there is a noticeable variance between the ice color, shape, and texture, which may require constant re-tuning of classical segmentation solutions, making them less robust for the application. In contrast, artificial neural networks (ANN) provide a more robust solution to the segmentation problem with its ability to train for varying conditions while capturing both the global and local context of an image, resulting in significantly improved results for changing environmental factions. The incremental training capability of ANNs is heavily utilized to improve machine vision front ends of popular industrial systems such as autonomous driving.

Deep learning neural networks have become the established approaches for modern semantic segmentation problems. There is a wide array of neural network architectures for semantic segmentation with varying accuracy and speed trade-offs intended for different application domains. The basic network architectures consist of convolutional layers connected such that a final pixel-wise prediction to segment the image is produced, e.g., fully connected convolutional neural network (FCN) [8]. SegNet [9] presents another popular architecture which uses an encoder-decoder network to label the image. Depending on the task that the neural networks are built for, these networks become increasingly complex. U-Net [10] presents an expansion of the FCN, adding skip connections and symmetry between the encoder and decoder layers to yield better accuracies. Additional layers in architectures and different activation functions further increase the effectiveness of the network for segmentation problems. A layer can be defined as a collection of nodes or neurons mapping the inputs of a network to its output. A deep neural network is a network with several layers that handle increasingly complex input-output relationships. Work in [11] presents a deep network with 16-19 layers to process an image. A recent development termed PSPNet [12] proposes a method of parsing the image post convolutional neural networks (CNN) to improve image, yielding excellent results with the trade-off of network complexity and efficiency. With the wide variety of networks available for semantic segmentation, selecting a functional and feasible architecture for a given application is a challenge.

The unavailability of training datasets poses a second challenge to the creation of a semantic segmentation network for sea ice classification. For established deep learning applications there are readily available datasets such as for autonomous driving [13], pedestrian tracking [14], and object recognition [15]. However, to the best of authors' knowledge, there are no labeled datasets available in literature for the purpose of classifying ice types in polar environments. In such scenarios, custom labeled datasets are created to meet the study's needs, which is the approach adopted in this work. Automotive applications reported in [16] creates a custom dataset for vehicle classification based on camera position, and [17] develop a large dataset of 14,144 images for vehicle classification. Marine applications reported in [18] design its own custom dataset for coral reef segmentation, and work in [19] design a custom dataset for oceanic Eddies. The coral reef dataset has a size of 413 images, where the authors perform nine class classification of the data. Typically dataset sizes of the order of 100's are utilized in these studies while larger, more representative datasets would offer enhanced performance of the networks. To remedy the smaller datasets that occur as a result of a custom dataset, there are two common approaches; data augmentation and transfer learning. Data augmentation serves to artificially expand the dataset through various image transformations or adjustments to create new images for training. Commonly used effective data augmentation methods are simple geometric transformations, horizontal flipping, color space modifications, and random cropping [20]. Transfer learning [21] is a technique used to improve the learning of a new neural network through the transfer of pre-trained model weights. In transferring the weights, the new model is able to start the training process using the previous knowledge and adapting it to the new dataset. This leads to reduced training times and potential higher accuracies in the model quicker.

Camera placement and configurations can also be used to improve the sea ice classification. In the work presented in this paper, a singular GOPRO camera is used to capture the imagery. While a single camera offers valuable visual information to the classification network, there is a lack of depth information in the imagery. As presented in [22], the usage of multiple cameras at different locations allows for

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depth information to be extracted from the imagery, which can provide additional information for classification purposes.

To the best of authors' knowledge, this work is the first work in its field using in-situ onboard camera imagery to classify sea ice using neural networks. Closely related applications to this work include [23], where authors present a semantic segmentation CNN to classify river ice from aerial drone footage of rivers. Additionally, [24] uses a CNN to classify various ice objects. However, instead of classifying ice types, this work focuses on ice objects such as icebergs and large ice floes. Work in [25] proposes the usage of sentinel-1 synthetic aperture radar data to classify sea ice using neural networks, where satellite imagery is used instead of onboard camera imagery for sea ice classification. Lastly, [26] proposes an improved ice navigation system with the combination of ship-based lidar, satellite imagery, and onboard visual classifications from operators. Sea ice detection and classification in operational settings are carried out by an ice services specialist considering several information sources, including ice radar, any available satellite, aerial imagery, and in-situ visual identification. The concentration of different ice types in a given zone directly correlates to the operational risk of the vessel governed by International Maritime Organization (IMO) standard specifications. Furthermore, the details of the ice concentration of different zones serve as additional inputs to the daily ice charting activities of the international ice patrol and the Canadian ice service [27]. This work targets to automate the in-situ visual detection and classification of ice to support both operational risk assessment and ice charting activities onboard vessels. The authors have previously investigated traditional image processing-based methods [28] [29], which have resulted in systems that are sensitive to lighting conditions, while not a having a structured means of improving the system based on expert feedback. This work presents the use of deep learning for the purpose of both detection and classification of sea ice with above 90% success rate for a large (1100 images, ~559 million pixels) dataset representative of varying environmental conditions. In the process, the authors have successfully evaluated transfer learning using the "cityscapes" dataset [30] for the application to overcome the limitations related to not having sufficient data for training. The system results in inference speeds of few seconds that are adequate to support onboard in-situ maritime ice navigation scenarios. The use of a deep learning procedure further enables re-training as new labeled data from the field is incorporated into the system.

As commencing work in the area of sea ice image segmentation, this paper presents a dataset creation, training, and evaluation of deep neural networks for sea ice detection and classification. In order to first assess the feasibility of a deep learning approach for sea ice imagery, a manually labeled training dataset is created from the Nathaniel B. Palmer Antarctic expedition imagery [31], i.e., the sea ice detection dataset. We analyze two different deep-learning-based semantic segmentation algorithms, perform transfer learning, and compare their performance in this task using accuracy metrics. Upon verifying feasibility, we expand the work to be able to classify different ice types in sea ice imagery using a manually labeled 1,090 image dataset, i.e., the sea ice classification dataset.

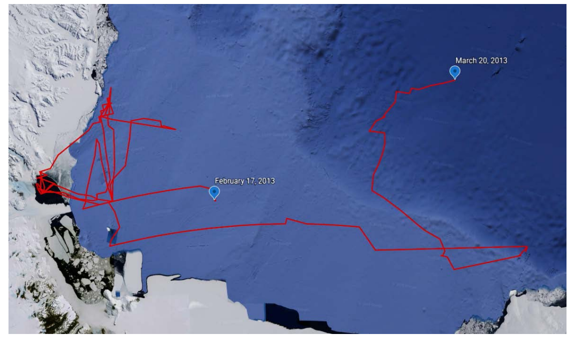

Fig. 2. The route of the Nathaniel B. Palmer from February 17th, 2013 to March 20th in the Ross Sea.

In this paper, we make the following contributions.

- We produce two novel datasets of labeled images from aboard an icebreaker in polar seas for deep learning sea ice detection and classification studies. First is the sea ice detection dataset, which includes 240 labeled images and four classes. Second is the sea ice classification dataset, which includes 1,090 labeled images and four additional classes capturing the type of ice seen in the images.
- Establish the detailed benchmark performance for the datasets using the SegNet and PSPNet semantic segmentation architectures, with transfer learning using the Citscapes dataset [30]. The study was capable of achieving an accuracy of 97.8% for sea ice detection and IoUs of 76%, 82%, and 93% for classification of new, grey, and first-year ice types respectively.

II. Dataset Creation

In this study, we first investigate the feasibility of deep semantic segmentation for sea ice detection using images. This is performed using the sea ice detection dataset presented in Section II.A. Upon successful feasibility assessment, we perform a detailed evaluation of deep neural networks for different sea ice type classification using images. This is performed using the sea ice classification dataset presented in Section II.B. Both datasets were created using footage from an actual polar icebreaker journey.

A. Sea Ice Detection Dataset

The first dataset is a multi-class dataset constructed from Go-pro images taken from the Nathaniel B. Palmer ice breaker on its two-month expedition through the Ross Sea, Antarctica. Video footage of the dataset can be seen in https://youtu.be/BNZu1uxNvlo. This dataset is comprised of 720p and 4K high definition images taken from a fixed location on the icebreaker. The images were periodically captured throughout each of the days, with upwards of 1600 photos captured each day. Each of the days presents an array of different conditions encountered in the voyage ranging from midday sun to gray skies and the setting sun. In addition, some days present precipitation on the lenses, which obscure the overall image. The purpose of this dataset is to evaluate if semantic segmentation is feasible for ice detection

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aboard icebreakers. This set is designed to be simplistic in nature, with only four classes being: Ice, Vessel, Ocean, and Sky. This dataset is drawn from four unique days of the journey containing different ice coverage and types, lighting, and environmental conditions commonly encountered in polar voyages. Approximately 60 photos were chosen from each day and labeled into the four classes using the freely available watershed labeling tool PixelAnnotationTool [32]. The dataset was then broken up into three sets, being training, validation, and testing. The data was split into each of the categories with 80% for training, 10% for validation, and 10% for testing. Due to the limited size of the training set, data augmentation was applied to the images via horizontal mirroring to give a final training dataset of 382 images.

B. Sea Ice Classification Dataset

The second dataset used in this study is also extracted from the same Nathaniel B. Palmer go-pro imagery, as described above. Unlike the last dataset, there is an increased focus on ice types and a variety of images used in the set. Of the entire available images, every day was used in the creation, with the exception of days that only contained night-time footage. This decision to exclude the night-time images from the set is due to the small sample size and limited visibility of the actual ocean state from the camera location. This creates a broad dataset capturing a large portion of the environmental conditions and ice types encountered on the icebreaker journey, resulting in a large variance of ice types in the dataset. For each of the days in the dataset, approximately 60 images were chosen at random to be labeled. Each of the images has a corresponding pixel-wise label, breaking the image down into the eight distinct classes. The pixel-wise labels were created using careful and detailed labeling with the same watershed labeling tool used above [32].

The training dataset consists of 896 images with approximately 53 images from each of the days. Each of the images has a corresponding pixel-wise label, breaking each image down into the eight distinct classes. For validation, approximately six images were randomly selected from each of the 17 days of footage to give 99 images for validating the model. The test set is comprised of 102 images selected from the 15 days used for training and an additional two days that were not used for the training and validation sets.

C. Dataset Class Labels

The semantic segmentation dataset considers eight classes, four classes to describe the non-ice objects, and four classes for ice types. The non-ice objects are defined as follows:

- Ocean: all open water in the image.
- Vessel: all sections of the boat, including the mast, people, and flag visible on the front of the boat.
- Sky: all visible sky in the image.
- Lens Artifacts: any particles or objects on the camera lens obstructing the image.

The model is designed to find and label all pixels to each of the classes with a focus on separating the ocean and different ice classes. The proper classification of the ice is crucial as certain mislabeled ice would defeat the purpose of the entire labeling and would provide little relevant data to any navigation or mapping system. A labeled image from this data model showcasing the majority of the classes can be seen in Fig. 3. It is important to note that the labeled image is colorized for visualization purposes, and the actual input labeled images are grayscale ranging from 0-7. There are several characteristics related to sea ice considered when creating "egg codes" for ice charts [33]. The egg code of an ice charts reports the types of ice (based on thickness), concentration (using a scale of 10), and form of ice (based on the size of ice floes). This information is also used in the IMO Polar Operational Limit Assessment Risk Indexing System (POLARIS), where concentrations related to nine different ice types are used for creating the ice numeral risk rating as given in Table I of [34]. Accurate classification of these types requires an ice service specialist and other onboard sea ice observation aids [35]. As this work is basing ice type classification only on sea ice imagery and labeling is performed by a knowledgeable non-expert user, only four visually distinguishable ice types are used in this work. The non-expert labeler consulted an expert for classification validation and used reference material [34] and [36] to follow established visual ice classification procedures when labeling the images. The considered sea ice types are as follows.

Fig. 3. An example of a labelled training image showing the five classes: the boat in red, ocean in dark blue, sky in cyan, firs-year ice in white, grey ice in grey-blue, and new ice in pale blue.

TABLE I
THE PROPERTIES OF THE NATHANIEL B. PALMER IMAGES

Number of images	42,947
Image type	jpg
Image size	1920x 1440
Days	26
Frequency of capture	30 photos per second
Camera info	GoPro HD2
Camera Height	17m from the water line

1) New Ice: New ice is one of the difficult ice types to classify. It is used as a general term for recently formed ice which, includes frazil ice, grease ice, and shuga. While these ice types vary in looks, they all are thin and weak formations of ice, which are loosely frozen together. In addition to the classical new ice, nilas is also included as new ice in this dataset. Nilas is a thin elastic crust of ice with a matte crust and up to 10 cm thick. While this is not new ice, the similarity in appearance and equivalent weakness of new ice make it a valid inclusion as New ice for classification. New ice is predominantly seen as smooth patches of water in the dataset with very little color. New ice is seen in Fig. 4(c).

2) Grey Ice: Grey ice, in the classification, represents two classes of ice being both grey ice and grey-white ice. The main difference between these two types of ice is that for grey ice,

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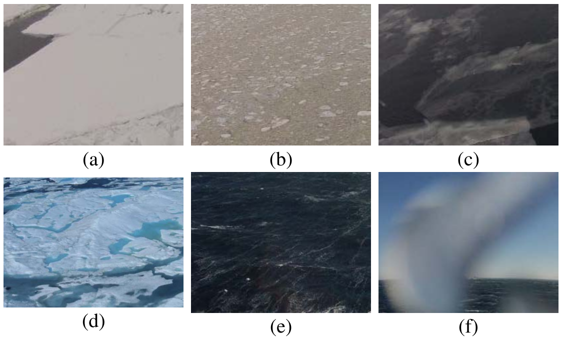

Fig. 4. Examples of (a) first-year ice, (b) grey ice, (c) new ice, (d) multiyear ice, [36], (e) ocean, (f) lens artifact from the dataset.

the thickness is 10-15 cm thick, whereas grey-white ice is young ice 15-30 cm thick. These classes have been combined to simplify the classification and the inability to calculate the thickness of the ice based on the current dataset. The grey color is the primary visual indicator for this ice; however, the ice shape and overall texture of the ice are additional clues to identify the ice. An example of grey ice is seen in Fig. 4(b).

3) First-Year Ice: First-year ice is described as floating ice of up to one year of growth and developed from young ice. The ice thickness ranges from 0.3 to 2 meters in thickness and is commonly seen as pack ice and sheets in the Arctic. In standards, this ice classification is broken down into additional categories based upon the thickness being:

- Thin First-year Ice/White Ice - first stage: 30-50 cm thick,
- Thin First-year Ice/White Ice - second stage: 50-70 cm thick,
- Medium First-year Ice: 70-120 cm thick,
- Thick First-year Ice: Greater than 120 cm thick.

However, due to the difficulty determining the ice thickness based purely off a fixed camera front-facing view, the ice has been all grouped into a single class. The main visual features used to detect and classify the ice for labelling is the solid white color and typical flat surface. To simplify the labelling process, the broken pieces of pans in rubble fields have also been defined as First-year ice. In the dataset, it appears as both ice pans and sheets of ice. An example of first-year ice is seen in Fig. 4(a).

4) Multiyear Ice: The last classification of ice for labelling is multiyear ice. This type of ice is old ice that has seen at least two summer's melt. The visual distinction of this ice is the smooth hummocks with large interconnecting, irregular puddles on top of the ice. It has a blue color to it where bare. Unfortunately, in the Nathaniel B. Palmer imagery, there does not exist any multiyear ice due to it being an Antarctic voyage. Any additional expansions to the dataset will aim to include this ice classification as it is a critical type of ice to distinguish between. An example of multiyear ice is seen in Fig. 4(d).

III. Network Architecture Selection

The neural network architectures analyzed for this project were chosen due to their success and popularity in the semantic segmentation field. Each architecture presents benefits and disadvantages with varying performances.

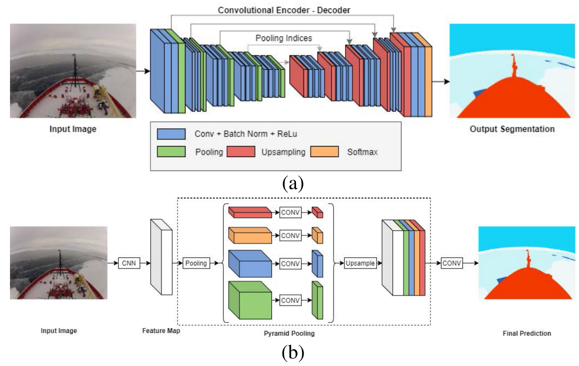

Fig. 5. (a) The structure of the SegNet encoder-decoder architecture [9]. (b) The architecture of the PSPNet model, in the case of the PSPNet101 model implemented, the CNN input seen in the figure is the ResNet101 model [12].

A. SegNet

Segmentation Network (SegNet) [9] is a deep encoder-decoder architecture developed by the University of Cambridge. This model features a quick and relatively simple neural network to perform semantic segmentation. It is offering a significantly less intensive training time and processing time compared to more complex network architectures. The neural network functions through using a sequence of non-linear processing layers known as encoders and a corresponding set of decoders with a pixel-wise classifier to label the images. Additionally, each encoder consists of one or more convolutional layers, followed by non-overlapping max-pooling and subsampling layers. Through the max-pooling indices in the decoders, it allows the network to retain high-frequency information such as shapes and contextual relationships while reducing the overall number of trainable parameters in the decoders [9].

B. PSPNet101

Pyramid Scene Parsing Network (PSPNet) was introduced for semantic segmentation to provide the capability of global context information to increase the accuracy of the classifications. This model provides a high level of accuracy compared to all other models seen in the field; however, it requires an intensive and lengthy training process due to a large number of parameters. PSPNet101 functions through the combination of the ResNet101 convolutional neural network and a four-layer pyramid pooling module to create the output results [12]. The four-layer pyramid pooling allows the network to develop different region-based context aggregation with division into different sub-regions in the images. The ResNet101 CNN is implemented at the start of the pyramid scene parsing to extract features. In the case of ResNet101, it contains 101 layers that use skip connections in the neural network that serves to prevent significant visual feature loss across the layers. This skip connection implemented is responsible for the models' ability to learn the classification identity without losing context or residual learning in which it obtained its name [37].

IV. Training

To train the detection networks, the SegNet model was trained from scratch using a windows machine with an Intel

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i5-4670K CPU and 16 GB of RAM. There were 382 images in our training set, resized to 918x688. The model was trained across five epochs with the training performance evaluated after each iteration to determine if additional training was required. The PSPNet101 network was trained using transfer learning for the cityscapes 713x713 dataset. Due to the models' heavy training process, a virtual machine with 16 vCPU and 60 GB of RAM was created and trained for nine epochs across the 382 training images resized to 713x713. For both of these models, due to the small dataset size, there was no batch size, and all images were used in each epoch. Additionally, the smaller epoch numbers were used to give the overall performance of both architectures to evaluate each model's performance.

For the ice classification dataset, to train both neural networks, a virtual machine on google cloud compute was created to optimize training. Both models were trained using a Linux based, 16 vCPU with 60 GB of RAM. Although CPU was used to train the model due to ease of access, in future work, GPU will be used to expedite the training process for large datasets. For the SegNet model, it was trained for 100 epochs with a batch size of 512 images. The network was trained from scratch as no pre-trained models were readily available in the model implementation. For the PSPNet model, the network was trained using transfer learning from the cityscapes dataset. Due to the transfer learning, the model was trained with 713x713 images. For both of the models, the intersection over union (IOU) and pixel-wise accuracy was computed after each epoch. The final weights used for both models were the models with the best performance across the validation set. In Fig. 6, the pixel-wise accuracies of validation sets for both architectures on both datasets are seen. It is noted that the PSPNet101 model has fewer epochs due to the intensive training time (2 hours per epoch). For both models, it is observed that the training set accuracy is higher than validation as expected; however, it is noted that neither model started to overfit at their max epoch. For the SegNet model, the best performance was determined to be at epoch 72, while the best performance for PSPNet101 is at the 24th epoch.

Both networks for semantic segmentation for icebreakers were evaluated using four performance metrics:

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$FP\ rate(FPR) = \frac{FP}{FP + TN} \tag{2}$$

$$FN\ rate(FNR) = \frac{FN}{FN + TP} \tag{3}$$

$$IoU = \frac{area\ of\ intersection}{area\ of\ union} \tag{4}$$

Where given a class (A), positives are the pixels that are predicted as belonging to that class. Negatives are the pixels that are predicted as not belonging to that class. True positive (TP): correct positive prediction; False positive (FP): incorrect positive prediction; True negative (TN): correct negative prediction; False negative (FN): incorrect negative prediction. IoU (Intersection Over Union) is a measure of how well the bounding boxes fit the actual location of an object.

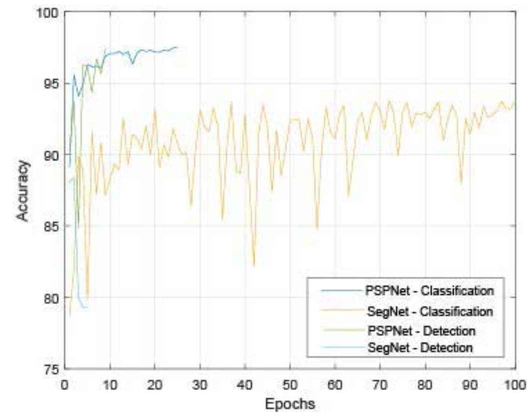

Fig. 6. Pixel-wise accuracy for the validation sets of both neural networks for the ice classification dataset and the ice detection dataset.

Where intersection and union are the intersection and union of the true and predicted pixel bounding boxes.

V. Results

The results below were obtained on the validation dataset, which was created to provide a range of polar icebreaker operational conditions. These images are sourced from the same days in which the neural network was trained upon however, they differ from the images used for the training data. Where the images are sourced from an actual image feed of an icebreaker, they provide an excellent metric on how the network would perform in the field. While this network only provides the results for day images using one vessel, it showcases the potential and overall performance of semantic segmentation for ice classification.

A. Performance on Sea Ice Detection Dataset

The performance of both models for sea ice detection is presented in Table II. PSPNet101 outperformed SegNet in every metric with an average IoU of 90.1% compared to 69.8% for SegNet, demonstrating a much greater contextual awareness. The pixel-wise accuracies of the models present that the SegNet architecture has significantly more classification errors as no class is over 95% compared to PSPNet, which boasts all classes at least 96% accurate. When analyzing the actual classification results in Fig. 7a-d, the SegNet results have large patches of misclassifications and struggle with differentiation between classes with similar coloring. Additionally, any large variation, including water droplets on the lens and midday glare off the ice, causes errors. For PSPNet101, there are few misclassifications visible in the results, with the model performing well across the variety of polar environments. The PSPNet101 neural network's actual performance as it is run across a section of day's journey can be found in the following video link https://youtu.be/Otx2Pu0XUdM.

B. Performance on Sea Ice Classification Dataset

Table II presents the results for the SegNet and PSPNet architectures for the sea ice detection dataset. SegNet was only able to achieve an average IoU of 53.6%, while the model performed well for the vessel and sky classifications; it only reached an IoU of 54.8% for the combined ice types. PSPNet101 had a higher average IoU at 75.5% with a combined ice classification IOU of 83.8%. For every classification

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TABLE II

IoU, Pixel-Wise Accuracy, False Positive Rate, False Negative Rate Performance of the PSPNet101 and SegNet Deep Neural Networks on Both Datasets. The Ice Class Represents a Combination of the Accuracies for Each Ice Type in the Classification Dataset. Since the Detection Dataset Does Not Have Some of the Classes of the More Detailed Dataset, They Are Left Blank

Metric	Network	Average	Ocean	Vessel	Sky	New Ice	First-Year Ice	Grey Ice	Lens Artifacts	Ice
IoU	SegNet (Classification)	53.6	66.9	96.9	91.2	31.9	73.5	59.1	9.0	54.8
IoU	**PSPNet101 (Classification)**	**75.5**	**89.9**	**98.9**	**99.0**	**76.1**	**93.4**	**81.9**	**65.0**	**83.8**
IoU	SegNet (Detection)	69.8	44.1	80.4	74.7					80.0
IoU	PSPNet101 (Detection)	90.1	80.8	89.3	95.7					94.7
Pixel-wise	SegNet (Classification)	96.7	96.9	99.3	96.7	95.3	92.9	96.4	99.7	94.9
Pixel-wise	PSPNet101 (Classification)	99.2	99.2	99.7	99.7	98.5	98.3	98.8	99.9	98.6
Pixel-wise	SegNet (Detection)	92.0	89.5	95.5	89.3					93.6
Pixel-wise	PSPNet101 (Detection)	97.8	96.6	97.7	98.4					98.5
FN Rate	SegNet (Classification)	28.0	16.6	1.5	3.2	59.3	20.8	4.1	90.4	28.1
FN Rate	PSPNet101 (Classification)	7.7	8.4	0.5	0.5	12.4	4.6	2.6	25.3	6.5
FP Rate	SegNet (Classification)	2.0	2.1	0.5	3.4	1.6	2.7	3.6	0.0	2.6
FP Rate	PSPNet101 (Classification)	0.5	0.1	0.2	0.3	0.9	0.7	1.1	0.0	0.9

IoU, PSPNet presents a higher accuracy. The pixel-wise accuracies between both models present a more promising result for SegNet. However, in each class it is marginally outperformed by PSPNet101. In Fig. 7, PSPNet101 demonstrates its strength across the varying conditions with no obvious errors in any of the results. SegNet, on the other hand, presents difficulties with varying environments. The classification of the vessel and sky is consistent however, there are clear errors in the results, including sky as ice due to the ship shadows and errors from the sun's reflection off the ocean. While the SegNet architecture has poor overall accuracy for segmentation, the PSPNet101 demonstrated excellent potential in the field. Overall, PSPNet101 had a higher accuracy of the two models. PSPNet101 has good average IoU performance, along with excellent IOU for all classes with the exception of new ice and lens artifacts. SegNet IoU demonstrates that while it had good pixel accuracy, the errors had a significant impact on the IoU, only resulting in an average IoU of generating 53.6% and having two classes perform poorly with individual IoUs of 31.9% and 9.0%. This disparity in network performance is not unexpected, as PSPNet101 is a significantly more complex and deep neural network than SegNet. The PSPnet101 architecture's performance across an entire day of images can be found at the following link https://youtu.be/LisIE47Kz28.

In both networks, they appear to struggle with the proper classification of new ice and lens artifacts. For the case of the lens artifacts, these appear as water droplets on the lens and have little representation in the dataset. Due to the labeling method, the overall region labeled as lens artifacts is difficult to define properly and overlap with the more common classifications. For the new ice, it confused most with first-year ice and the ocean class, as seen in Fig. 9. From the confusion matrix given in Fig. 9a, it is evident that the SegNet has a poor performance on the small classes, with a widespread of false positives. The new ice classification is commonly confused for first-year ice, grey ice, and ocean, and the lens artifacts are more often confused as both sky and ocean classes than its own class. The first-year ice is also occasionally confused with the sky, which is the case in Fig. 7 h(2) and k(2), where there are sections of the ice labeled as the sky. Lastly, the ocean class is seen to be mistaken as ice classes. For PSPNet, Fig. 9b has the same issue with new ice confused with grey ice and first-year ice; however, it is to a much smaller degree. There is little confusion observed between non-ice classes and ice classes. Lastly, the first-year ice and grey ice classes are occasionally confused with each other around 2.2% and 1.6%, respectively. However, this confusion, as seen in Fig. 7 is mainly caused by these classes commonly bordering on each other.

For SegNet, the rates of the FN and FP are typically higher than that of PSPNet101. There is an extremely high rate of false negative detection for new ice and lens artifacts. The ocean and first-year ice classes also are classified wrong, with only the vessel, grey ice, and sky classes having a low false negative rate. The false positive rate average for SegNet is only 2.0%, with grey ice, sky, and ocean commonly being mistaken. The lens artifacts class has a low FP rate due to the limited classification of the class in the model. For PSPNet, new ice and lens artifacts have the highest FN rate with new ice and grey ice having the highest FP rate. This is as expected because both new ice and lens artifacts have the most confusion between other classes. The vessel, sky, and first-year ice have minimal FN and FP rates.

Although there is no directly comparable work to benchmark the performance, we provide a comparison of our results with the following closely related applications. Work in [25] applies a custom lightweight CNN architecture using three convolutional layers, which achieves an overall pixelwise accuracy of 91.6% for a 77-image validation dataset. Similarly, [23] achieves a mean IOU of 88.1% and pixel accuracy of 95.9% using a custom CNN for 3-class segmentation. Finally, [24] uses the ResNet architecture to achieve an accuracy of 90% for a 39 image test dataset for ice object classification. The results achieved for detection and classification presented in Table II. is comparable with the performance of these related application domains which use CNNs. It is important to note that a higher complexity architecture than the networks used in [23]–[25] are needed for the proposed work due to the challenging application domain. The lightweight CNN architecture considered in this work, Segnet, significantly underperformed for ice class classification as shown in the results comparison given in Table II.

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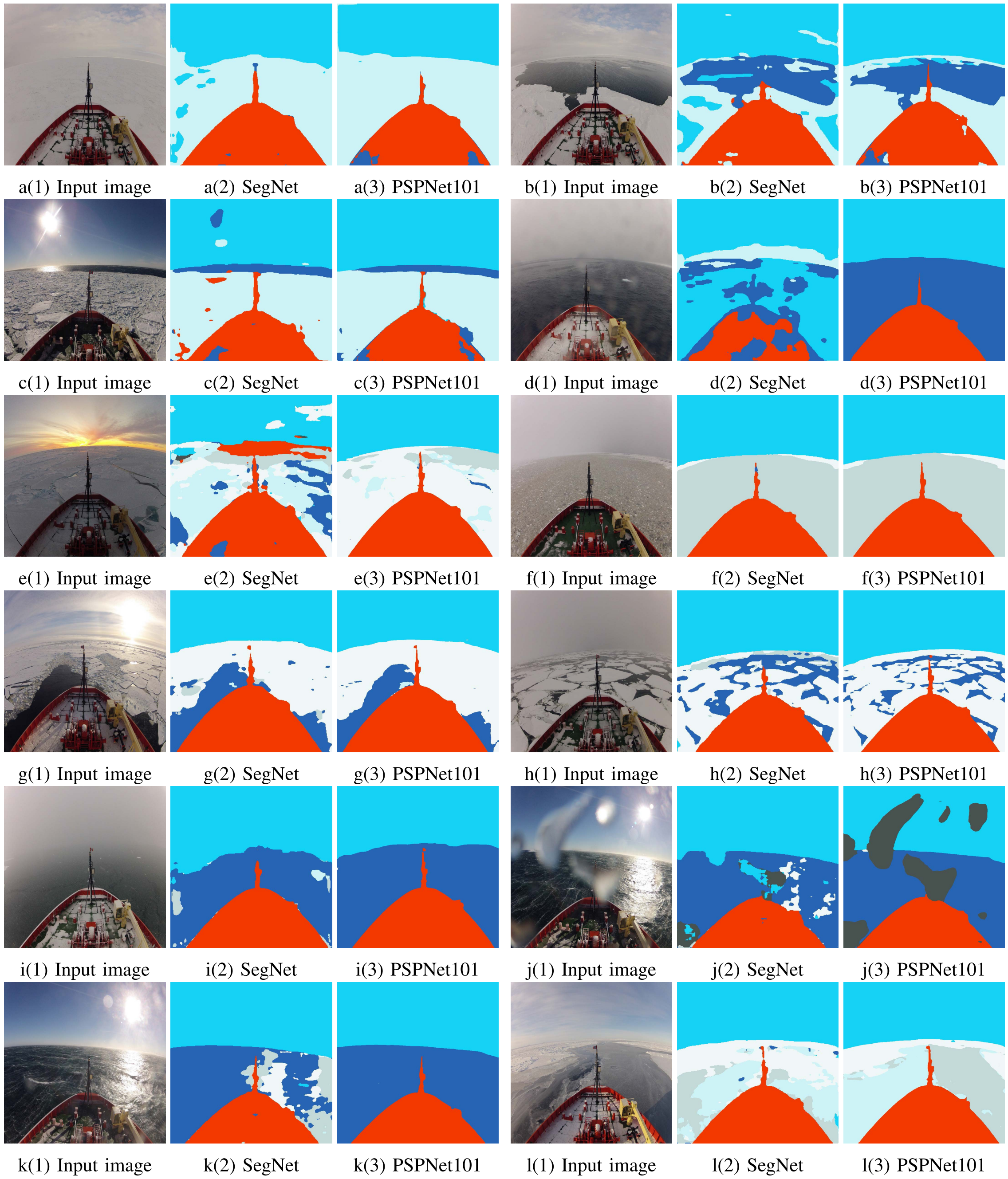

Fig. 7. (a)–(d) present the performance of the ice detection models using the test images. (e)–(l) presents a comparison of segmentation results of both ice classification models using test images. (1), (2), (3) are used to indicate the image, SegNet prediction of classes, and PSPnet prediction of classes respectively with the boat in red, the ocean in dark blue, the sky in cyan, first-year ice in white, grey ice in grey-blue, and new ice in pale blue. In images, both neural networks' overall segmentation performances are observed on images from the test set.

Fig. 8 illustrates the performance processing time trade-off between the two networks for both sea ice detection and classification. The trade-off between the models is apparent, where PSPNet presents a more accurate segmentation at the cost of speed. Between both datasets, there is little change in the frames per second of the PSPNet model, indicating that the number of classes in the classification dataset does not significantly impact the model speed.

In terms of which method is better suited for semantic segmentation of polar icebreakers, it is evident that PSPNet101 is the obvious choice from the standpoint of accuracy and performance. SegNet provides an underlying representation; however, it falls short to fully recognize the image's overall context and struggles with less represented classes such as lens artifacts and new ice. If the overall speed of the network were an important factor in the results, SegNet would be considered more. It is capable of passing multiple frames in a second as opposed to the long processing time per frame of the PSPNet101 model. However, it is important to note that given the slowly changing surroundings perceived in marine operational settings, the around 3-second processing time per image of PSPNet is quite

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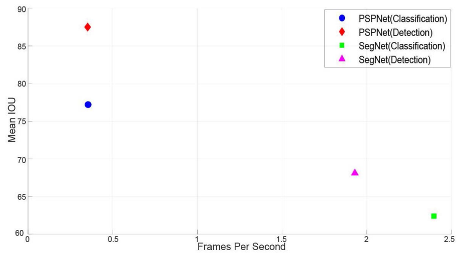
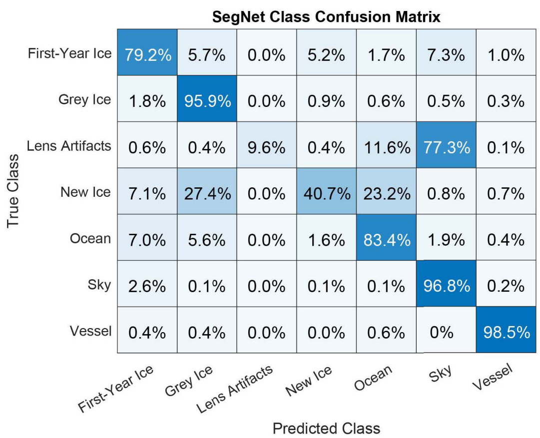
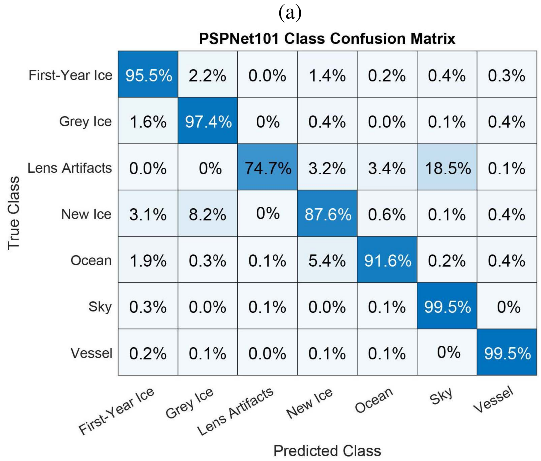


Fig. 8. Speed vs. mean IOU for both architectures for both datasets. A clear divide between accuracy and speed is presented between both models. The running time is measured on an AMD Ryzen 7 3800X with 32 GB of RAM.



(a)



(b)

Fig. 9. (a) The confusion matrix for the SegNet architecture on the ice classification dataset. (b) The confusion matrix for the PSPNet101 architecture on the ice classification dataset.

acceptable for a bridge-mounted ice monitoring and mapping system.

VI. Conclusion

In this paper, two different deep semantic segmentation networks were trained and evaluated on polar icebreaker environments to validate the performance of the methods for sea ice detection and classification. A dataset was built from imagery captured aboard the Nathaniel B. Palmer icebreaker and was devised a data model of multiple ice types, ocean, sky, lens artifacts, and vessel. Using this dataset, a SegNet and PSPNet101 model was trained and evaluated. The PSPNet101 model was trained using transfer learning techniques from a pre-trained model. The performance of both models was evaluated across accuracy and IoU acquiring metrics for both networks. Additionally, the confusion matrices for both networks were presented to highlight where the misclassifications arise. The results show that the PSPNet architecture can attain 97% accuracy for sea ice detection and achieves false negative rates of 12.4%, 2.6%, and 4.6% for new, gray, and first-year ice types classification, respectively.

In the future, the authors hope to expand this work by further increasing the training dataset size and incorporate data from different polar voyages. Increased segmentation of the detected first-year ice is desired to further meet the desired classifications. Additionally, detecting the freeboard of ice and rolling faces of ice are of crucial importance to realize automated navigation and mapping aids for arctic environments. Lastly, multiyear ice is desirable for the network to detect. However, in the current dataset, there are no cases present.

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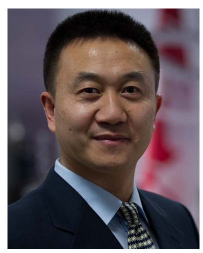

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